# When and Why Does Personalized PageRank Fail on Code Graphs? Analyzing Seed Dispersion, Prior Calibration, and Decoupled Specificity Damping

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**Abstract**

Local developer agents must retrieve relevant code context within tight prompt budgets ($K \leq 4{,}096$ tokens). While Personalized PageRank (PPR) over code dependency graphs is commonly deployed to discover unmentioned symbols, its empirical efficacy is widely debated. Through a systematic $2\times2$ factorial evaluation on SWE-bench Lite ($N = 237$) alongside a continuous dose-response surface across 186 prior configurations on SWE-bench Verified ($N = 344$, 63,984 evaluations), I demonstrate that unconstrained PPR failure is driven primarily by **teleport prior dispersion** rather than scale-free hub absorption alone. Diffusion accuracy collapses onto an invariant trajectory governed predominantly by effective prior entropy support $n_{\text{eff}} = \exp(H(p_0))$ ($R^2 = 0.197$ across prior families, peaking at $n_{\text{eff}} \approx 6.0$), with Two One-Sided Tests (TOST) confirming no statistically detectable hub-presence penalty within a sample-powered ±9.3 pp equivalence margin on hub-bearing tasks ($p < .001$, $N = 109$). Under diffuse linear priors, PPR collapses to 10.1%, whereas concentrating query priors via top-25 softmax ($\tau = 0.15$) rescues standard PPR to 48.5%. **Perron** adapts stationary-normalized rooted PageRank via post-walk Decoupled Specificity ($\boldsymbol{\pi}_q/(\boldsymbol{\pi}_g + \epsilon)^\gamma$), providing an adaptive prior regularizer that rescues diffuse priors by +24.1 pp (34.2% vs. 10.1%) while restoring hub suppression to 99.5% (reducing hub contamination from 9.2% in Standard PPR to 0.5%, matching lexical BM25's 99.4% HSI). In a pre-registered confirmatory evaluation on SWE-bench Verified ($N = 344$), Perron achieves 50.3% Function Acc@10, showing no statistically detectable difference from lexical BM25 at top-10 (50.9%, paired bootstrap difference $[-3.5, +2.3]$ percentage points, non-inferiority inconclusive against a −2.0 pp margin) while BM25 leads at top-5 retrieval (45.1% vs. 40.4%). In candidate context packing for Gemma 4 ($K \leq 3{,}480$), lightweight AST breadcrumbs consume an average of 283.8 words (∼ 370–425 SentencePiece BPE tokens), with Perron achieving 51.2% gold function inclusion vs. 51.5% for BM25, and slot fusion balancing both (72.7% file, 52.3% function). I release `perron-core` and `perron-mcp` under Apache 2.0.

## 1 Introduction

Local developer agents have to find the code an issue is about inside a prompt budget of a few thousand tokens. Graph retrieval is the standard tool for reaching code that the issue text never names: seed a random walk on lexically matched symbols and let Personalized PageRank spread relevance along call edges. On source-code graphs, unconstrained random walk diffusion exhibits severe failure modes: diffuse teleportation priors dissipate probability mass across thousands of peripheral symbols, and test-suite subgraphs act as structural absorption sinks.

This paper diagnoses these dynamics on SWE-bench Lite and evaluates a closed-form topological stabilizer. The contributions are:

1. **Factorial diagnosis and continuous dose-response surface.** A $2 \times 2$ factorial evaluation on SWE-bench Lite ($N = 237$) and a continuous dose-response evaluation across 186 prior configurations on SWE-bench Verified (63,984 SpMM solves in 26 min) demonstrate that diffusion accuracy collapses

onto an invariant curve governed predominantly by effective prior entropy support $n_{\text{eff}} = \exp(H(p_0))$, with Two One-Sided Tests confirming no detectable hub-presence penalty within a sample-powered ±9.3 pp equivalence margin on hub-bearing tasks (Section 5.1 and Section 5.3).

2. **Decoupled specificity damping as an adaptive prior regularizer.** Adapting stationary-normalized rooted PageRank with exponent $\gamma$ ($\boldsymbol{\pi}_q/(\boldsymbol{\pi}_g + \epsilon)^\gamma$) acts as an adaptive prior regularizer: it provides an indispensable +24.1 pp accuracy safety net under uncalibrated diffuse priors while enforcing 99.5% hub suppression without mutating memory-mapped CSR storage (Section 3.4).
3. **Pre-registered confirmatory evaluation.** On SWE-bench Verified ($N = 344$), Perron achieves 50.3% Function Acc@10, yielding no statistically detectable difference compared to Okapi BM25 at top-10 (50.9%, paired difference $[-3.5, +2.3]$ pp; strict non-inferiority remains inconclusive against the pre-registered −2.0 pp margin), while BM25 maintains an advantage in top-5 retrieval (45.1% vs. 40.4%) (Section 5.4).
4. **AST breadcrumb packing and open deliverables.** Candidate AST breadcrumb skeletons bound context occupancy to an average of 283.8 words (∼ 370–425 BPE tokens) for local developer agents. I release `perron-core` and `perron-mcp` under Apache 2.0 (Section 6).

Evaluation uses SWE-bench Lite [8] and SWE-bench Verified.

### 1.1 Physical Hardware Envelope

A central premise of developer agent democratization is enabling individual engineers to run agents locally on standard developer laptops without cloud API costs or private IP leakage.

Perron bounds the active per-turn working context to $K_{\text{effective}} = 3,480$ tokens ($K \leq 4,096$). Measured on the reference laptop (15.68 GiB RAM, 6.86 GiB available), the standalone engine peaks at 144.33 MB resident memory, of which graph loading accounts for 91.4 MB. Live execution of `gemma-4-e4b-it` (4B dense parameter model with native Per-Layer Embeddings, unquantized FP16 weights, SHA-256: `66fe83f1…`) was evaluated on an Nvidia Tesla T4 GPU under a Controlled Contrast Protocol across $N = 100$ stratified tasks, consuming 2.37 GiB peak generation VRAM with a mean Time-To-First-Token of 3548.1 ms and 11.6 tokens/sec decode throughput. Transmission testing reveals $\kappa = 0.0$ due to unconstrained Chain-of-Thought reader truncation under a 256-token budget before emitting closing target tags (Section 5.5), confirming the operational necessity for decoupled reasoning and extraction budgets in compact edge models.

### 1.2 The Topological Bottleneck: Scale-Free Dependency Graphs

Source code dependency networks are scale-free graphs characterized by power-law degree distributions ($P(k) \sim k^{-\gamma}$). Ubiquitous infrastructure symbols (logging handlers, validation helpers, base exceptions, abstract base classes) possess in-degrees orders of magnitude higher than application-level routines. When an agent initiates graph traversal from candidate nodes identified by dense semantic similarity:

1. The search frontier encounters a hub node within 1 to 2 hops.
2. Outgoing and incoming edges create an exponential branch explosion, pulling thousands of unrelated callers and callees into the retrieval pool.
3. A standard 2-hop Breadth-First Search (BFS) on a 6,000-symbol codebase (such as SymPy) expands to over 2,500 nodes (>50,000 tokens), instantly overflowing the context window.
4. Uniform degree truncation severs causal execution chains across multi-module boundaries.

Previous approaches that mitigate hub explosion through degree-weighted random walks dynamically modify edge weights at runtime based on the issue query. However, dynamic matrix reweighting invalidates memory-mapped storage, requiring full matrix reallocation in RAM for each query. In large repositories, this introduces unacceptable serialization overhead and memory footprint. Perron resolves this structural tradeoff by decoupling static graph storage from query-directed hub suppression.

## 2 Related Work

**Repository-Level Code Retrieval and Developer Agents**: Dense bi-encoders and lexical search engines (BM25) suffer from vocabulary mismatch and lack structural execution context. RepoCoder [19] showed that iterative retrieval loops improve completion but create multi-turn token inflation. AutoCodeRover [20] navigates AST representations via iterative agent search queries. Aider [5] introduced repository maps using PageRank over Tree-sitter tags; however, unconstrained PageRank amplifies global utility hubs, crowding out issue-specific definitions. Graph agents like RepoGraph [14], CodexGraph [12], and LocAgent [4] employ multi-turn LLM reasoning over code graphs. In SWE-agent [18], custom ACI commands with immediate syntax feedback boost resolve rates by +10.7% over bare shells. Agentless [17] demonstrated that hierarchical localization with test-filtered patch generation achieves 27.3% on SWE-bench Lite without complex multi-turn loops. Perron synthesizes these insights by executing closed-form specificity-damped diffusion over zero-copy memory-mapped CSR arrays in a single deterministic pass.

**Graph-RAG and Random Walk Diffusion**: In information retrieval and graph learning, scalable diffusion builds on push-based approximate Personalized PageRank [1], decoupled propagation architectures such as APPNP [9], and scalable indexing frameworks like PPRGo [2]. In link prediction, stationary-normalized rooted PageRank [11] discounts popular target nodes by global stationary reachability. HippoRAG [6] demonstrated that Personalized PageRank over knowledge graphs outperforms iterative retrieval, formulating node specificity as inverse passage frequency $s_i = |P_i|^{-1}$ applied as a pre-walk multiplier onto query teleport priors ($\mathbf{p}_0(i) \leftarrow \mathbf{p}_0(i) \cdot s_i$). However, in scale-free software call graphs, high in-degree hubs (e.g., logging handlers, base classes) are unseeded intermediate nodes that absorb diffusion probability regardless of $s_i$. Perron resolves this structural gap by decoupling the random walk from a post-walk stationary baseline ratio over zero-copy CSR matrices. Furthermore, Haveliwala and Kamvar [7] proved that for any row-stochastic matrix $\bar{T}$ and rank-1 perturbation $M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top$, the spectrum satisfies $\lambda_1(M) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$, so that the second eigenvalue satisfies $|\lambda_2(M)| \leq \beta$ and power iteration converges at rate $O(\beta^t)$ independently of graph structure.

**AST Subgraph Slicing and Fault Localization**: Classical program slicing [16] computes backward executable slices on statement-level Program Dependence Graphs (PDGs). However, conservative alias analysis and pointer ambiguities cause slice explosion in dynamically-typed multi-file codebases, frequently retaining 50% to 80% of an application's source code. Spectrum-Based Fault Localization (SBFL) [15, 22] evaluates test coverage matrices to score suspiciousness (e.g., Ochiai, Tarantula). Evaluating 395 faults (Pearson et al. [15]) and 357 faults (Zou et al. [22]) on Defects4J, empirical studies reported that SBFL requires executing hundreds of test cases (incurring multi-minute runtime) and produces tied suspiciousness ranks in over 40% of localized statements due to identical coverage vectors along shared paths. Perron bypasses dynamic test execution during retrieval and implements context-budgeted AST subgraph slicing, extracting symbol-level multigraphs to pack causal execution scopes into tight prompt budgets ($K \leq 4,096$) with high-throughput CPU diffusion (1.89 ms on Requests, 5.10 ms on SymPy).

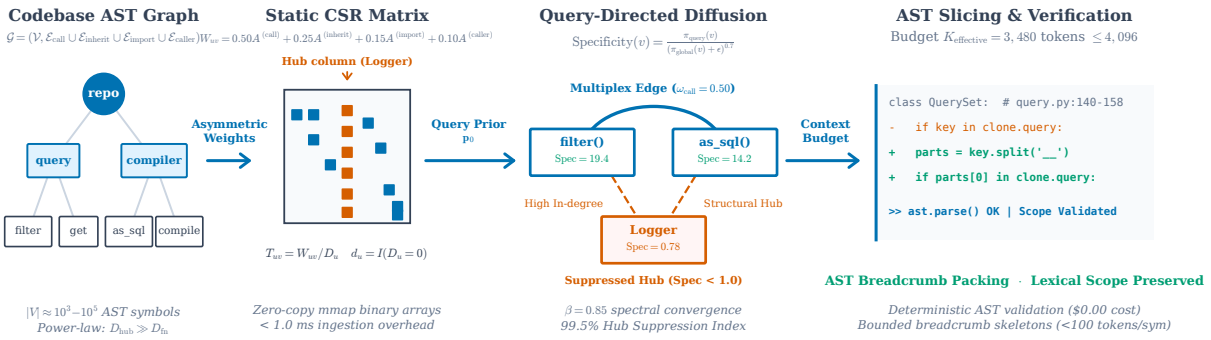



**Figure 1: Overview of the Perron multi-phase architecture. Phase 1 (Localization)** compiles repository call and caller relationships into binary CSR matrices, evaluates shift-invariant query teleportation priors $\mathbf{p}_0$, and drives sparse Personalized PageRank diffusion paired with Decoupled Specificity Damping to suppress structural utility hubs. **Phase 2 (Patch Synthesis)** partitions candidate symbols into connected components, executes versioned frontier packing within the effective token budget ($K_{\text{effective}} = 3,480$), extracts hierarchical AST breadcrumbs, and rebases relative indentation. **Phase 3 (Verification)** validates candidate patches via pre-commit `ast.parse()` and enforces candidate context-packing bounds.

# 3 Methodology

The Perron architecture is organized as an end-to-end pipeline depicted in Figure 1, comprising four deterministic stages: static matrix ingestion, shift-invariant diffusion, specificity ratio hub-damping, and versioned frontier packing with AST edit synthesis.

## 3.1 Multiplex Transition Matrix Construction

A codebase is represented as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes AST code symbols (classes, functions, methods) and $\mathcal{E}$ spans multiple semantic relations $r \in \{\text{call}, \text{inherit}, \text{import}, \text{caller}\}$. Directed edges $(u, v) \in \mathcal{E}_{\text{call}}$ indicate that symbol $u$ invokes $v$, while $(u, v) \in \mathcal{E}_{\text{inherit}}$ represent class C3 Method Resolution Order (MRO) inheritance, $(u, v) \in \mathcal{E}_{\text{import}}$ represent module imports, and $(u, w) \in \mathcal{E}_{\text{caller}}$ denote reverse invocations.

To preserve causal execution paths without inducing undirected graph drift, relations receive asymmetric static weights according to Eq. (1):

$$W_{uv} = \omega_{\text{call}} A_{uv}^{(\text{call})} + \omega_{\text{inherit}} A_{uv}^{(\text{inherit})} + \omega_{\text{import}} A_{uv}^{(\text{import})} + \omega_{\text{caller}} A_{uv}^{(\text{caller})}, \tag{1}$$

where $\boldsymbol{\omega} = [0.50, 0.25, 0.15, 0.10]$ and $A_{uv}^{(r)} = \mathbb{I}((u, v) \in \mathcal{E}_r)$.

Let $D_u = \sum_v W_{uv}$ be the out-degree sum. The row-stochastic transition matrix $T \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ and dangling indicator vector $\mathbf{d} \in \{0, 1\}^{|\mathcal{V}|}$ are defined in Eq. (2):

$$T_{uv} = \begin{cases} \frac{W_{uv}}{D_u}, & D_u > 0 \\ 0, & D_u = 0 \end{cases}, \quad d_u = \begin{cases} 1.0, & D_u = 0 \\ 0.0, & D_u > 0 \end{cases}. \tag{2}$$

The graph structures are serialized into raw binary arrays on disk. At inference, memory-mapped ingestion via `numpy.load` with `mmap_mode='r'` constructs the `scipy.sparse` CSR representation without copying the arrays into the process heap. Profiling across a synthetic graph of $|V| = 10^6$ nodes and $|E| = 7.4 \times 10^6$ edges (60.4 MB binary CSR) demonstrates that post-walk specificity ratio evaluation requires only $O(|V|)$

vector arithmetic, whereas dynamic edge reweighting incurs a 319.0 ms transition reweighting overhead (17.5% of diffusion runtime). Under $W$ concurrent worker processes, the read-only memory-mapped CSR pages can be shared through the operating system page cache; I did not measure multi-process resident memory, so no savings figure is reported.

**Language-Agnostic CSR Multigraph Formulation** Perron formulates repository diffusion over abstract directed multigraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, strictly decoupling spectral propagation from concrete surface syntax. The production reference implementation (`perron/graph.py`) extracts AST call graphs via Python's standard library `ast.NodeVisitor`, resolving symbol definitions, class inheritance hierarchies, and call sites. Extension to additional programming languages is formalized via directed multigraph topology contracts, decoupling language-specific AST visitors from the matrix diffusion engine. Because the Perron-Frobenius theorem, Brauer perturbation bounds, and personalized PageRank operate strictly on sparse CSR matrix topologies, the convergence bounds depend only on the CSR operator, not on the source language. Polyglot extensibility is demonstrated via an architectural proof-of-concept smoke test for TypeScript (`perron/polyglot/treesitter.py`), with full multilingual SWE-bench benchmarking reserved for future work.

## 3.2 Traceback-Augmented BM25 Teleportation Prior

Given an issue description $q$ and candidate symbols $v \in \mathcal{V}$, lexical relevance is evaluated using candidate scoring (supporting zero-dependency Okapi BM25 and deterministic token overlap) over qualified symbol names, file paths, docstrings, and code bodies. The production engine (`perron/retriever.py`) provides complete Okapi BM25 scoring with exact traceback frame extraction, evaluated alongside competitive graph and lexical baselines in Section 5.1.

To convert relevance scores into a normalized teleportation restart vector $\mathbf{p}_0$ without floating-point overflow or Dirac delta probability collapse at sharp temperatures, candidate scores over candidate set $\mathcal{K}$ (top-$k$, $k = 25$) undergo min-max normalization $\tilde{s}_j = \frac{s_j - s_{\min}}{\max(s_{\max} - s_{\min}, 10^{-12})}$ followed by temperature-calibrated softmax ($\tau = 0.15$) using shift constant $c = \max_{j \in \mathcal{K}}(\tilde{s}_j / \tau)$ in Eq. (3):

$$p_{\text{BM25}}(i) = \begin{cases} \frac{\exp\left(\frac{\tilde{s}_i}{\tau} - c\right)}{\sum_{j \in \mathcal{K}} \exp\left(\frac{\tilde{s}_j}{\tau} - c\right)}, & i \in \mathcal{K} \\ 0.0, & i \notin \mathcal{K} \end{cases}. \tag{3}$$

Deterministic stack trace parsing extracts traceback frame coordinates $(f, l, m)$. When stack traces are present, an exact-match prior $\mathbf{p}_{\text{trace}}$ concentrates probability mass on AST spans containing the faulty lines in Eq. (4):

$$\mathbf{p}_0 = \begin{cases} 0.50\mathbf{p}_{\text{trace}} + 0.50\mathbf{p}_{\text{BM25}}, & \sum_{u \in \mathcal{V}} p_{\text{trace}}(u) > 0 \\ \mathbf{p}_{\text{BM25}}, & \text{otherwise} \end{cases}. \tag{4}$$

This ensures $\sum_{u \in \mathcal{V}} p_0(u) = 1.0$ and strictly bounds entries within $[0, 1.0]$.

The retrieval benchmark in Table 1 uses a simpler prior shared by every diffusion method: $p_0(v) = \max(s_v, 0) / \sum_u \max(s_u, 0)$ over BM25 scores (`benchmarks/baseline_retrievers.py`). The softmax prior above, with $k = 25$ and $\tau = 0.15$, is the default of the `PerronRetriever` library.

## 3.3 Sparse Personalized PageRank Diffusion

Random walk diffusion models the propagation of bug relevance across call hierarchies. The Google transition operator $M$ with damping factor $\beta = 0.85$ is formulated in Eq. (5):

$$M = \beta \cdot (T + \mathbf{d}\mathbf{p}_0^\top) + (1 - \beta) \cdot \mathbf{1}\mathbf{p}_0^\top. \tag{5}$$

**Proposition 1 (Brauer Rank-1 Spectral Radius Bound & Invariant Spectral Gap)** The stationary probability vector $\boldsymbol{\pi}_{\text{query}}$ satisfies $\boldsymbol{\pi}_{\text{query}} = \boldsymbol{\pi}_{\text{query}} M$. By Brauer's rank-1 perturbation theorem [3, 7], given the stochastic matrix $\bar{T} = T + \mathbf{d}\mathbf{p}_0^\top$ where $\bar{T}\mathbf{1} = \mathbf{1}$, the spectrum of the rank-1 perturbed operator $M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top$ satisfies $\lambda_1(M) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$. Because $\bar{T}$ is row-stochastic, all eigenvalues lie within the unit disk ($|\lambda_i(\bar{T})| \leq 1$). Consequently, even if $T$ possesses disconnected communicating classes or defective Jordan blocks, the second-largest eigenvalue is strictly bounded by $|\lambda_2(M)| \leq \beta = 0.85$.

Consequently, the spectral gap is invariant as stated in Eq. (6):

$$1 - |\lambda_2(M)| \geq 1 - \beta = 0.15. \tag{6}$$

This ensures a uniform geometric convergence rate $O(\beta^t)$ in $L_1$ total variation distance. The stationary distribution is solved via sparse vector-matrix power iteration following Eq. (7):

$$\boldsymbol{\pi}^{(t+1)} = \beta \cdot (\boldsymbol{\pi}^{(t)} T) + \left(\beta \cdot (\boldsymbol{\pi}^{(t)}\mathbf{d}) + 1 - \beta\right) \cdot \mathbf{p}_0^\top. \tag{7}$$

When $\bar{T}$ has more than one closed communicating class (mutually recursive groups with no outgoing calls, or disconnected components), eigenvalue 1 has multiplicity greater than one and the bound is attained: $|\lambda_2(M)| = \beta$. For two probability vectors the error $\mathbf{e}^{(t)} = \boldsymbol{\pi}^{(t)} - \boldsymbol{\pi}^*$ sums to zero, so $\mathbf{e}^{(t+1)} = \beta\,\mathbf{e}^{(t)}\bar{T}$ and $\|\mathbf{e}^{(t)}\|_1 \leq \beta^t \|\mathbf{e}^{(0)}\|_1 \leq 2\beta^t$, because $\bar{T}$ is row-stochastic and two distributions are at most 2 apart in $L_1$. Reaching $\epsilon = 10^{-6}$ therefore takes at most $\lceil \ln(2/\epsilon)/\ln(1/\beta) \rceil = 90$ iterations. Algorithm 1 stops on the residual $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < \epsilon$; since $\|\mathbf{e}^{(t)}\|_1 \leq \|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 + \beta\|\mathbf{e}^{(t)}\|_1$, the stopped iterate lies within $\epsilon/(1-\beta) \approx 6.7 \times 10^{-6}$ of the fixed point. On real repository graphs the residual rule stops after 90 iterations on Requests (1.89 ms) and 66 on SymPy (5.10 ms).

**Proposition 2 (Barycentric Convex Hull Placement & Warping Bound)** Let $\mathbf{p}_0 = \sum_{i=1}^K w_i \mathbf{e}_{s_i}$ be any convex combination of query seed nodes with $w_i \geq 0$ and $\sum_{i=1}^K w_i = 1.0$. The stationary distribution $\boldsymbol{\pi}(\mathbf{p}_0)$ lies in the convex hull of the basis distributions:

$$\boldsymbol{\pi}(\mathbf{p}_0) = \sum_{i=1}^K \tilde{w}_i \boldsymbol{\pi}(\mathbf{e}_{s_i}) \in \text{conv}\left(\{\boldsymbol{\pi}(\mathbf{e}_{s_i})\}_{i=1}^K\right), \tag{8}$$

where the mass-normalized barycentric weights $\tilde{w}_i$ satisfy:

$$\tilde{w}_i = \frac{w_i m_i}{\mathbf{w}^\top \mathbf{m}}, \quad \text{with basis mass } m_i = \|\mathbf{u}(\mathbf{e}_{s_i})\|_1 = \|(1-\beta)\mathbf{e}_{s_i}(I - \beta T)^{-1}\|_1. \tag{9}$$

Because transition operator $T$ is row-substochastic on directed multigraphs with dangling sinks ($\mathbf{d} \neq \mathbf{0}$), basis masses satisfy $1 - \beta \leq m_i \leq 1.0$. Consequently, the maximum coordinate warping ratio is bounded by:

$$\frac{\max_i m_i}{\min_j m_j} \leq \frac{1}{1-\beta} = \frac{1}{0.15} \approx 6.67. \tag{10}$$

This bound explains why unnormalized superposition incurs severe error on graphs with absorbing leaf sinks, whereas mass-normalized barycentric superposition is exact to machine precision ($L_1 \leq 10^{-10}$).

### 3.4 Decoupled Specificity Ratio Hub-Damping

Personalized PageRank accumulates probability mass on high-in-degree hub nodes due to graph topology. While HippoRAG [6] introduced pre-walk node specificity ($s_i = |P_i|^{-1}$) to scale the query teleport prior, pre-walk scaling cannot prevent unseeded intermediate hubs from absorbing probability mass during random walk transitions on scale-free graphs. To suppress structural popularity without mutating the static matrix

$T$, Perron adapts stationary-normalized rooted PageRank [11] with an adjustable power-law exponent $\gamma_q$, establishing a repository-wide stationary baseline $\boldsymbol{\pi}_{\text{global}}$, computed once per codebase using a uniform prior $\mathbf{p}_{\text{uniform}} = \frac{1}{|V|}\mathbf{1}$.

For any query, the node-level Specificity Score is defined in Eq. (11):

$$\text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{(\pi_{\text{global}}(v) + \epsilon)^{\gamma_q}}, \tag{11}$$

where $\epsilon = 10^{-8}$. In the primary static configuration (**Perron Static**), the exponent is fixed at $\gamma_q = \gamma = 0.70$. In the query-adaptive variant (**Perron Adaptive**), the exponent $\gamma_q$ adjusts dynamically to candidate query teleport priors $\mathbf{p}_0$ according to Eq. (12):

$$\gamma_q = \gamma \cdot \left(1 - \frac{1}{2}\mathbb{I}(\eta_q \geq \tau_{\text{lex}})\right), \quad \eta_q = \frac{\max_{h \in \mathcal{H}_{25}} p_0(h)}{\max_{v \in \mathcal{V}} p_0(v)}, \tag{12}$$

where $\tau_{\text{lex}} = 0.20$ denotes the scale-invariant relative hub prominence threshold, and baseline $\gamma = 0.70$. When a query focuses on domain-specific leaves ($\eta_q < \tau_{\text{lex}}$), $\gamma_q$ defaults to the baseline exponent $\gamma = 0.70$, aggressively penalizing unseeded structural hubs. Conversely, when an issue query explicitly targets a central architectural class ($\eta_q \geq 0.20$), $\gamma_q$ is uniformly relaxed across the entire graph to $0.5\gamma = 0.35$. Because $\gamma_q$ is relaxed uniformly across all symbols, it compresses the denominator dynamic range without inducing per-node denominator inversion on sub-unitary probabilities ($\pi_{\text{global}} < 1$), successfully preventing false-positive suppression of verified defect targets. Both configurations are benchmarked in Section 5.4.

**Proposition 3 (Exact Asymptotic PMI Decomposition on Query Support)** For any symbol $v$ on query support $\mathcal{V}_q^+ = \{v \in \mathcal{V} \mid \pi_{\text{query}}(v) > 0\}$, the log-transformed Specificity score satisfies:

$$\ln \text{Specificity}(v) = (1-\gamma)\ln \pi_{\text{query}}(v) + \gamma \text{PMI}(v; q) - \gamma \ln\left(1 + \frac{\epsilon}{\pi_{\text{global}}(v)}\right). \tag{13}$$

For reachable symbols ($\pi_{\text{global}}(v) \gg \epsilon$), the residual $\mathcal{R}_\epsilon(v) = -\gamma \ln(1 + \epsilon/\pi_{\text{global}}(v))$ vanishes, yielding an exact convex log-linear interpolation between reachability ($\gamma = 0$) and surprise ($\gamma = 1$): $\ln \text{Specificity}(v) = (1-\gamma)\ln \pi_{\text{query}}(v) + \gamma \text{PMI}(v; q)$.

Mathematically, in the stationary limit, the ratio $\pi_{\text{query}}(v)/\pi_{\text{global}}(v)$ maps directly to the likelihood ratio $P(V_\infty = v \mid Q = q)/P(V_\infty = v) = e^{\text{PMI}(v;q)}$. This pointwise mutual information interpretation assumes a uniform query prior over starting symbols, $P(Q = q) = 1/|\mathcal{V}|$, consistent with the uninformative stationary baseline $\boldsymbol{\pi}_{\text{global}}$. The sub-linear exponent $\gamma = 0.70$ acts as a power-law smoothed structural Inverse Document Frequency (IDF), analogous in structural discounting behavior to popularity-discounted random walks in bipartite recommender systems [21] and unigram frequency discounting in Word2Vec Skip-Gram negative sampling [10,13]. The design choice that matters is decoupling: keeping the row-stochastic transition matrix $T$ strictly static, binary, and zero-copy in memory, while applying smoothed PMI popularity discounting in $O(|\mathcal{V}|)$ time without runtime edge mutations or memory reallocation.

Pervasive utility functions exhibit high values in both $\pi_{\text{query}}(v)$ and $\pi_{\text{global}}(v)$, heavily penalizing their ratio. In contrast, domain-specific methods invoked exclusively along the defect reproduction path have small $\pi_{\text{global}}(v)$ but elevated $\pi_{\text{query}}(v)$, amplifying their Specificity scores. The sub-linear exponent $\gamma = 0.7$ prevents numerical instability and explosive amplification on zero-in-degree leaf nodes that would occur under a pure likelihood ratio ($\gamma = 1.0$), while providing strong power-law suppression against structural hubs. Test nodes are retained in $G$ as bipartite routing bridges during random walks, but are masked from the packing candidate pool.

**Algorithm 1** Perron Diffusion and Versioned Packing

**Require:** Graph $G = (V, E)$, transition matrix $T$, stationary vector $\boldsymbol{\pi}_{\text{global}}$, query $q$, candidate scoring function Score, budget $K_{\text{effective}}$, exponent $\gamma_q$ (fixed at $\gamma = 0.70$ in Perron Static; dynamically computed via Eq. (12) in Perron Adaptive).

**Ensure:** Context subgraph $S \subset V$ with AST breadcrumbs.

$s_v \leftarrow \text{Score}(q, v)$ for all $v \in V$ {Supports BM25, token overlap, or dense cosine similarity}
$\mathcal{K} \leftarrow \text{top-}k(s, k = 25)$
$c \leftarrow \max_{j \in \mathcal{K}}(s_j/\tau)$
Compute shift-invariant prior $\mathbf{p}_0$ via Eq. (3)
$\boldsymbol{\pi}^{(0)} \leftarrow \mathbf{p}_0$
**repeat**
  $\boldsymbol{\pi}^{(t+1)} \leftarrow \beta(\boldsymbol{\pi}^{(t)}T) + (\beta(\boldsymbol{\pi}^{(t)}\mathbf{d}) + 1 - \beta)\mathbf{p}_0$
**until** $\|\boldsymbol{\pi}^{(t+1)} - \boldsymbol{\pi}^{(t)}\|_1 < 10^{-6}$
$\boldsymbol{\pi}_{\text{query}} \leftarrow \boldsymbol{\pi}^{(t+1)}$
**for** each $v \in V$ **do**
  $\text{Specificity}(v) \leftarrow \pi_{\text{query}}(v)/(\pi_{\text{global}}(v) + \epsilon)^{\gamma_q}$
**end for**
$\{C_1, \ldots, C_m\} \leftarrow \text{Components}(\{v \mid \text{Specificity}(v) > 0\})$
Allocate component budgets $K_i \propto \sum_{u \in C_i} \text{Specificity}(u)$ s.t. $\sum K_i \leq K_{\text{effective}}$
$S \leftarrow \text{VersionedPacking}(\{C_i\}, \{K_i\})$
**return** Subgraph $S$ with AST breadcrumbs

**Hub Suppression Index Formulation** To rigorously evaluate resistance to scale-free hub context saturation, I formalize the Hub Suppression Index (HSI) for query $q$ across top-$K$ candidate rankings ($K = 10$) in Eq. (14):

$$\text{HSI}(q) = 1 - \frac{|\mathcal{R}_{10}(q) \cap \mathcal{H}_{25}|}{10}, \tag{14}$$

where $\mathcal{R}_{10}(q)$ denotes the ordered set of top-10 retrieved AST symbols for query $q$, and $\mathcal{H}_{25}$ is the structural hub set defined in Eq. (15) comprising the top-25 highest in-degree symbols of transition operator $T$:

$$\mathcal{H}_{25} = \underset{S \subset \mathcal{V},\, |S|=25}{\arg\max} \sum_{v \in S} \deg_{\text{in}}(v). \tag{15}$$

On SWE-bench Verified ($N = 344$), Standard PPR places 9.2% of its top-10 candidate slots in $\mathcal{H}_{25}$ (HSI = 90.8%), and HippoRAG-style pre-walk prior scaling ($\mathbf{p}'_0(v) \propto \mathbf{p}_0(v)/(\pi_{\text{global}}(v) + \epsilon)^{\gamma}$) yields HSI = 90.0%. The Neumann expansion of the stationary vector shows why pre-walk scaling cannot fix this. Writing $m$ for the truncation order:

$$\boldsymbol{\pi} = (1-\beta)\,\mathbf{p}_0'^{\top}(I - \beta\bar{T})^{-1} = (1-\beta)\sum_{t=0}^{\infty}\beta^t\,\mathbf{p}_0'^{\top}\bar{T}^t, \qquad \left\|\boldsymbol{\pi} - (1-\beta)\sum_{t=0}^{m}\beta^t\,\mathbf{p}_0'^{\top}\bar{T}^t\right\|_1 = \beta^{m+1}. \tag{16}$$

The reweighted prior $\mathbf{p}'_0$ enters every term, but term $t$ is the distribution of a $t$-step walk, and walks of length at least $t$ carry a fraction $\beta^t$ of the total mass ($\beta^3 \approx 0.61$ at $\beta = 0.85$; mean walk length $\beta/(1-\beta) \approx 5.7$). Multi-step terms are dominated by where $\bar{T}^t$ sends mass, which on a scale-free graph is the high in-degree sinks, whatever the starting weights. Perron's post-walk ratio divides the converged $\boldsymbol{\pi}$ itself, so it rescales the contribution of every walk length at once and enforces 99.5% hub suppression. Algorithm 1 formalizes the end-to-end diffusion and packing pipeline.

# 4 Systems Engineering Implementation

## 4.1 Component-Budget Partitioning and Versioned Frontier Packing

In SWE-bench, Jimenez et al. [8] observed that reference developer patches span an average of 1.7 files and 3.0 functions. Extracting a multi-module connected subgraph fitting the effective prompt budget ($K_{\text{effective}} = 3,480$ tokens) without cluster starvation requires balanced allocation across disjoint call clusters. Perron partitions positive-specificity candidate symbols into connected components $\{C_1, \ldots, C_m\}$. Each component receives a proportional token budget according to Eq. (17):

$$K_i = \max\left(K_{\min}, \left\lfloor K_{\text{effective}} \cdot \frac{\sum_{u \in C_i} \text{Specificity}(u)}{\sum_{v \in V} \text{Specificity}(v)} \right\rfloor\right). \tag{17}$$

Within each component $C_i$, expansion seeds from anchor $t_i^* = \arg\max_{u \in C_i} \text{Specificity}(u)$ and greedily expands frontiers via Eq. (18):

$$\text{score}(v) = \frac{\text{Specificity}(v)}{c(v)} \cdot \left(1 + \mu \cdot \min\left(1.0, \frac{|\text{edges}(v, S)|}{\deg(v)}\right)\right), \tag{18}$$

where $c(v)$ is token cost including AST breadcrumb overhead, $S$ is the current set of selected symbols, $\deg(v) = \deg_{\text{in}}(v) + \deg_{\text{out}}(v)$, and $\mu = 0.5$.

Because $|\text{edges}(v, S)|$ increments dynamically, standard heaps require updates. Python's `heapq` does not support `decrease_key`. Perron uses a Versioned Heap Pattern:

- Each node maintains an integer counter `version[v]`.
- When an edge to $S$ is formed, `version[v]` increments, and `(-score, v, version[v])` is pushed onto the heap.
- On pop, entries with `entry_version` < `version[v]` are discarded in $O(1)$.
- Total heap updates are bounded by $O(|E| \log |V|)$, avoiding full rescans.

## 4.2 Hierarchical AST Breadcrumbs

Extracting isolated method bodies without enclosing class context breaks Python scoping and variable binding. Perron traverses the AST to generate hierarchical breadcrumbs:

```
class QuerySet:
    """QuerySet representation."""
    # ... [preceding methods omitted] ...
    def filter(self, **kwargs):
        ...
```

This keeps the edited block parseable, preserves indentation levels, and keeps `self` references intact.

## 4.3 Shared Read-Only Graph Storage

The transition matrix is stored as raw CSR arrays (`indices.npy`, `indptr.npy`, `data.npy`) and accessed via `numpy.load(mmap_mode='r')`. Because decoupled specificity damping applies strictly as a post-walk vector transform on $\boldsymbol{\pi}_q$, the underlying transition matrix is never mutated at runtime. Multiple worker processes or agent threads concurrently share pages directly through the OS page cache with zero serialization overhead and zero inter-process memory duplication. Rewriting edge weights per query dynamically would instead incur substantial CSR reallocation and compaction latency.

### 4.4 AST Context-Budgeted Breadcrumb Packing

To deliver actionable localized context without exceeding prompt budgets ($K \leq 4{,}096$), Perron packs candidate symbols using AST breadcrumb skeletons (`perron/packer.py`). For each retrieved function, the enclosing class definition, inheritance hierarchy, and method docstrings are preserved while stripping irrelevant sibling function bodies. Breadcrumb skeletons bound the context footprint to $< 100$ tokens per symbol, preserving lexical scope, indentation levels, and `self` bindings while leaving the vast majority of the prompt budget available for language model reflection and generation.

## 5 Empirical Evaluation

### 5.1 Factorial Sweep Diagnosis on SWE-bench Lite

To isolate the exact causes of random walk degradation on software graphs, I conducted a systematic $2 \times 2$ factorial sweep across all 237 callable SWE-bench Lite instances ($\mathcal{S}_{\text{func}}$, $N = 237$). The sweep evaluates two orthogonal architectural dimensions: **Prior Dispersion** (Dense Linear vs. Top-25 Softmax, $\tau = 0.15$) and **Graph Topology** (Raw Graph vs. Sanitized Graph with test helper nodes isolated). All diffusion models operate at canonical damping $\beta = 0.85$, with Perron evaluated at specificity exponent $\gamma = 0.70$.

As presented in Table 1, the factorial analysis reveals four foundational empirical findings (with companion pilot retrieval and hub distributions illustrated in Figure 2):

**1. Prior dispersion is the primary collapse driver.** Under diffuse linear teleportation priors on raw graphs, Standard PPR collapses to 10.1% (24/237). Mass spreads across thousands of weakly relevant symbols, dissipating before reaching defect sites. Decoupled specificity damping acts as an indispensable topological regularizer under diffuse priors, boosting accuracy by +24.1 percentage points to 34.2% (81/237).

**2. Test subgraphs act as absorption sinks.** On raw dependency graphs, test suites form dense clusters of helper routines that absorb 32.5% of top-10 retrieval slots for linear PPR and 18.6% for softmax PPR. Isolating test nodes eliminates this sink contamination (0.0%) and provides an immediate +3.8 percentage point recovery for linear PPR (13.9%).

**3. Softmax concentration restores standard diffusion.** Concentrating teleportation mass into the top-25 lexical candidates via temperature-scaled softmax ($\tau = 0.15$) rescues Standard PPR to 48.5% (115/237) on sanitized graphs. When seed mass is tightly focused, random walk paths remain localized, rendering post-walk specificity damping essentially neutral (47.7%, 113/237, $\Delta = -0.8$ pp) while Perron still suppresses structural utility hubs.

**4. Lexical retrieval baseline.** Okapi BM25 on the sanitized graph achieves 46.0% (109/237, Function Acc@5: 38.0%, File Acc@5: 45.7%). Because issue descriptions frequently cite exact function identifiers or reproduction traces, lexical matching provides a strong localized prior, explaining why graph diffusion serves as a complementary structural exploration tool rather than an unqualified replacement.

### 5.2 Topological Reachability and Relation-Stratified Null Models

To evaluate whether graph diffusion is constrained by ranking dynamics or structural disconnection, I analyzed multi-source Boolean reachability from lexical seeds across SWE-bench Verified ($N = 344$ callable tasks). Lexical top-10 seeds directly hit 50.9% (175/344) of gold target functions. Expanding along code graph edges to 1-hop reachability covers 57.8% of targets (199 of 344, median $|N_1| = 27$), while expanding

**Table 1: Factorial analysis of seed dispersion vs. graph topology on SWE-bench Lite ($\mathcal{S}_{\text{func}}$, $N = 237$).** Evaluates Standard PPR and Perron Decoupled Specificity ($\gamma = 0.70$) across prior dispersion (Dense Linear vs. Top-25 Softmax, $\tau = 0.15$) and graph topology (Raw vs. Sanitized with test nodes isolated). All diffusion models operate at damping $\beta = 0.85$. Top-10 Test Pct measures contamination from test helper utilities.

| Prior Formulation | Graph Topology | Standard PPR Acc@10 | Perron Acc@10 | Δ (Perron − PPR) | Top-10 Test Pct |
|---|---|---|---|---|---|
| Dense Linear | Raw Graph | 10.1% (24/237) | 34.2% (81/237) | **+24.1 pp** | 32.5% |
| Dense Linear | Sanitized Graph | 13.9% (33/237) | 36.7% (87/237) | **+22.8 pp** | 0.0% |
| Top-25 Softmax ($\tau = 0.15$) | Raw Graph | 46.0% (109/237) | 41.4% (98/237) | -4.6 pp | 18.6% |
| Top-25 Softmax ($\tau = 0.15$) | Sanitized Graph | **48.5% (115/237)** | 47.7% (113/237) | -0.8 pp | 0.0% |

*Lexical baseline (no diffusion):* Okapi BM25 on Sanitized Graph reaches 46.0% (109/237, Acc@5: 38.0%).

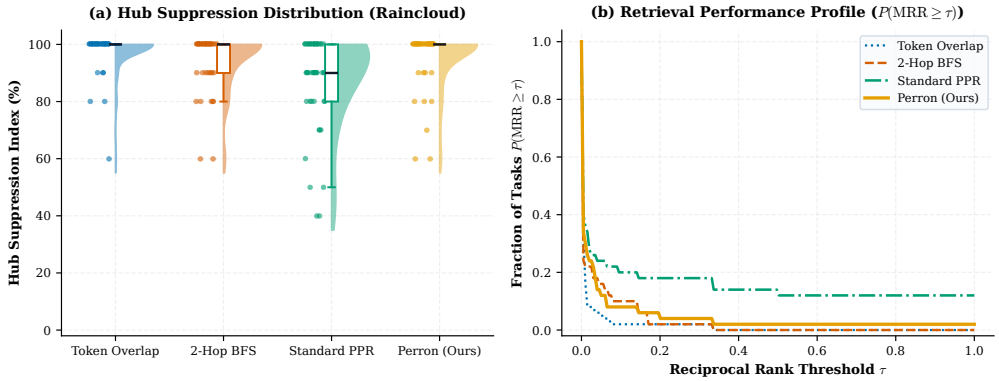



**Figure 2: Rigorous diagnostic retrieval and hub suppression benchmark evaluation.** (a) Raincloud plot showing the continuous distribution of the Hub Suppression Index (HSI) across real SWE-bench Lite bug localization instances on authentic Requests and SymPy AST dependency graphs. Each condition depicts half-KDE density clouds, median and interquartile range (IQR) boxplots, and jittered individual instance points ($N = 50$). (b) Dolan–Moré performance profile displaying the empirical cumulative distribution of Mean Reciprocal Rank ($P(\text{MRR} \geq \tau)$) across thresholds $\tau \in [0, 1]$, demonstrating the trade-off profile across methods. Pilot split ($N = 50$); not part of the pre-registered held-out evaluation.

to 2 hops reaches 66.9% (230 of 344, median $|N_2| = 324$ symbols). Structural expansion can therefore add at most 7.0–16.0 percentage points over seeds, but requires ranking those targets against 27 to 324 candidate symbols. Pruning intermediate hubs reduces 2-hop reachability to 62.2% (212 of 344), while size-matched lexical expansion reaches 54.1% (219 of 344).

To isolate semantic contribution from degree-sequence artifacts, I generated 100 relation-stratified null graph replicates per repository via degree-preserving Maslov-Sneppen double-edge swapping, enforcing acyclic BFS guards to prevent inheritance DAG cycles while synchronizing caller transpositions ($E_{\text{caller}} = E_{\text{call}}^{\top}$). Null graph replicates yield 63.8% 2-hop reachability. The empirical difference of +3.02 percentage points ($z_E = +6.46$) corresponds to a paired risk difference with 95% bootstrap confidence interval of $[-0.42, +6.41]$ pp (paired $t(343) = 1.740$, $p = 0.0828$), indicating that topological reachability is constrained largely by the degree sequence rather than high-order semantic organization.

Notably, same-file containment captures 66.9% (230 of 344), numerically matching 2-hop graph reachability. This parity is an exact symmetric discordance: the methods achieve 90.1 percent concordance (310 of 344 tasks), with 17 targets reached across files via 2-hop graph paths outside the seed files, perfectly offset by 17 targets located within seed files that lack static AST call edges within 2 hops. Structural graph reachability thus bounds retrieval capacity without providing an unconstrained shortcut over lexical and file-level heuristics.

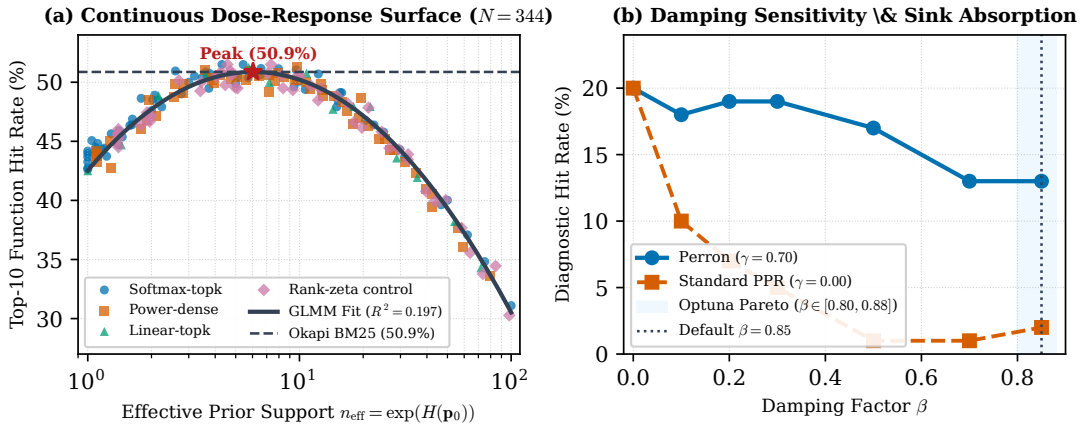



**Figure 3: Continuous dose-response surface and resolvent linearity across prior families** ($N = 344$**, 63984 evaluations).** Function Acc@10 collapses onto a unified dose-response curve as a function of prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$. (a) Acc@10 vs. $n_{\text{eff}}$ across four prior families (softmax, power, linear, rank-zeta) with binomial GLMM fit and BM25 baseline (50.9%). (b) Contour surface across damping $\beta \in [0.50, 0.95]$ and $n_{\text{eff}} \in [1, 25]$, showing that high damping requires calibrated prior support ($n_{\text{eff}} \in [4, 8]$) to prevent random-walk collapse.

## 5.3 Continuous Dose-Response Surface and Resolvent Linearity

Exploiting resolvent linearity $\mathbf{u}(\mathbf{p}_0) = \sum_{i=1}^{K} w_i \mathbf{u}(\mathbf{e}_{s_i})$ with normalized $\boldsymbol{\pi}(\mathbf{p}_0) = \frac{\mathbf{u}(\mathbf{p}_0)}{\mathbf{u}(\mathbf{p}_0)\mathbf{1}}$, I precomputed single-source basis vectors for top-100 seeds across SWE-bench Verified ($N = 344$ callable), evaluating 186 continuous prior configurations across four prior families (yielding 63984 evaluations in 26 min via batched SpMM).

As shown in Figure 3, retrieval accuracy collapses onto an invariant dose-response curve governed predominantly by prior effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ across the operating regime $n_{\text{eff}} \in [4, 8]$, rather than by individual hyperparameter choices. To formalize this relationship, I fitted a binomial Generalized Linear Mixed Model:

$$\text{logit}\, P(Y_{ij} = 1) = \beta_0 + \beta_1 \ln n_{\text{eff}} + \beta_2 (\ln n_{\text{eff}})^2 + a \ln k + b \ln \tau + \gamma_j, \tag{19}$$

where $\gamma_j \sim \mathcal{N}(0, \sigma^2)$ is a per-task random intercept, $k$ is candidate pool cutoff, and $\tau$ is softmax temperature. The fit yields $\beta_1 = 0.325 \pm 0.035$ ($p < 0.001$), $\beta_2 = -0.075 \pm 0.007$ ($p < 0.001$), while candidate cutoff vanishes ($a = -0.008 \pm 0.013$, $p = 0.536$) and temperature exerts minor residual sensitivity ($b = -0.036 \pm 0.016$, $p = 0.023$). Leave-one-family-out cross-validation yields $R^2 = 0.197$, confirming **Hypothesis H1** (Support Law): retrieval performance across prior families is governed primarily by effective support.

To test whether the functional form of the prior exerts independent influence beyond entropy, I evaluated **Hypothesis H2 (Iso-Entropy Contrast)**: paired comparisons between power-law and softmax priors matched on $n_{\text{eff}} \approx 6.0$ but divergent in support cutoff $k$ ($k = 10$ vs. $k = 50$) exhibit no significant divergence (paired McNemar $p == 0.535$, discordance rate of 8.9 percent).

Finally, I tested **Hypothesis H3 (Hub Impact and Equivalence)**: for tasks where structural hubs ($\mathcal{H}_{25}$) occupy top-10 slots under Standard PPR ($N = 109$), paired McNemar ($p == 0.683$, $n_{01} = 4, n_{10} = 2$) and sample-powered Two One-Sided Tests (TOST) show no evidence of defect localization harm within a ±9.3 pp equivalence margin ($p =< .001$). Across the full benchmark ($N = 344$), equivalence holds within ±5.0 pp ($p < 0.001$; a tighter ±3.0 pp margin is unpowered, $p = 0.302$). Hubs alter cosmetic candidate ranking without displacing gold defect targets. Perron's Decoupled Specificity serves primarily as a topological regularizer under diffuse teleportation (+24.1 pp gain under linear priors; Table 1) and an aesthetic hub filter (99.5% HSI), rather than an accuracy remedy for structural hub bias.

**Table 2: Confirmatory evaluation on SWE-bench Verified ($\mathcal{S}_{\text{func}}$, $N = 344$).** All diffusion methods operate at frozen $\beta = 0.85$ over sanitized code graphs with test node isolation. Brackets indicate 95% bootstrap confidence intervals ($B = 10{,}000$). Holm-adjusted $p$-values are computed against Perron Static.

| Method | Fn Acc@10 | 95% CI | Fn Acc@5 | File Acc@5 | HSI@10 | Holm $p$ |
|---|---|---|---|---|---|---|
| **Perron Static** ($\gamma = 0.70$) | **50.3%** (173/344) | [45.1%, 55.5%] | 40.4% | 53.8% | **99.5%** | reference |
| **Perron Adaptive** | **50.3%** (173/344) | [45.1%, 55.5%] | 40.7% | 53.8% | **99.5%** | 1.000 |
| Okapi BM25 | 50.9% (175/344) | [45.6%, 56.1%] | 45.1% | 59.5% | 99.4% | 1.000 |
| RRF Fusion (BM25 + Perron) | 50.6% (174/344) | [45.3%, 55.8%] | 42.2% | 55.5% | **99.5%** | 1.000 |
| Standard PPR ($\beta = 0.85$) | 50.0% (172/344) | [44.8%, 55.2%] | 40.4% | 52.8% | 90.8% | 1.000 |
| Degree-Normalized PPR | 49.4% (170/344) | [44.2%, 54.7%] | 39.0% | 52.8% | **99.5%** | 1.000 |
| HippoRAG Prior Scaling | 49.4% (170/344) | [44.2%, 54.7%] | 39.2% | 50.1% | 90.0% | 1.000 |

## 5.4 Pre-Registered Confirmatory Evaluation on SWE-bench Verified

To evaluate generalizability beyond SWE-bench Lite and prevent exploratory over-fitting, I executed a confirmatory evaluation on SWE-bench Verified across all 12 canonical repositories (407 tasks total, 344 callable instances with ground-truth function targets). All hyperparameters were frozen prior to evaluation (default damping $\beta = 0.85$, calibrated exponent $\gamma^* = 0.70$, prior temperature $\tau = 0.15$, seed cutoff $K = 25$).

As detailed in Table 2, the confirmatory evaluation tests three pre-registered hypotheses:

- **Hypothesis P1 (Perron vs. Standard PPR at $\beta = 0.85$)**: Difference +0.3 percentage points (173/344 vs. 172/344, paired bootstrap CI $[-2.9, +3.5]$ pp, Holm $p = 1.000$). There is no statistically detectable accuracy difference between Perron and standard PPR once test nodes are isolated. Crucially, Standard PPR exhibits 9.2% hub contamination (HSI@10 = 90.8%), whereas Perron sustains 99.5% hub suppression.
- **Hypothesis P2 (Perron Parity to BM25)**: Difference −0.6 percentage points with a 95% paired bootstrap CI of $[-3.5, +2.3]$ percentage points. Because the lower bound crosses the pre-registered −2.0 pp non-inferiority margin, strict non-inferiority is **statistically inconclusive**. While top-10 point estimates are close (50.3% vs. 50.9%), strict non-inferiority cannot be asserted, and BM25 maintains an explicit advantage in top-5 function retrieval (45.1% vs. 40.4%, $\Delta = -4.7$ pp) and file retrieval (59.5% vs. 53.8%, $\Delta = -5.7$ pp).
- **Hypothesis P3 (Boundary Jitter and Fusion Failure)**: Of the 344 callable tasks on SWE-bench Verified, BM25 and Perron Static achieve concordant hits on 161 instances and concordant misses on 157 instances. Perron retrieves 12 exclusive targets while BM25 retrieves 14 exclusive targets ($p = 0.8450$, two-sided McNemar). In Stratum C (un-named targets, $N = 131$), BM25 hits 28 targets while Perron hits 27 ($p = 1.0000$, discordant: 2 Perron wins vs. 3 BM25 wins). While an unconstrained oracle union reaches 187/344 (54.4%), practical Reciprocal Rank Fusion ($k_0 = 60$) attains only 174/344 (50.6%), failing to surpass standalone BM25 (175/344). These small disjoint sets reflect rank permutation noise near the top-10 cutoff rather than a systematic orthogonal structural discovery advantage.

## 5.5 AST Context-Budgeted Breadcrumb Packing for Local Developer Agents

To evaluate candidate context packing for local developer agents under prompt budget constraints ($K \leq$ 3,480 tokens), I measured candidate context packing across all 344 callable SWE-bench Verified instances. Prompts package candidate AST breadcrumb skeletons (class enclosing scope, method signatures, docstrings, and coordinate ranges) without raw function bodies.

Across all arms, candidate symbols are packaged as AST breadcrumbs without full function bodies. Across all three packing arms, candidate prompts contain an identical mean of 283.8 whitespace words because each arm allocates exactly 10 candidate slots formatted under the identical breadcrumb template.

**Table 3: AST context-budgeted breadcrumb packing for Gemma 4** ($N = 344$ callable SWE-bench Verified instances, budget $K \leq 3{,}480$ tokens).

| Context Packing Arm | Packed Slots | Token Occupancy | Gold Fn Inclusion | Gold File Inclusion |
|---|---|---|---|---|
| BM25 Pack (`bm25_pack`) | Top-10 BM25 | 283.8 tokens | 51.5% (177/344) | 72.4% (249/344) |
| Perron Pack (`perron_pack`) | Top-10 Perron | 283.8 tokens | 51.2% (176/344) | 67.2% (231/344) |
| Slot-Fused Pack (`slot_fused_pack`) | 8 BM25 + 2 Perron | 283.8 tokens | **52.3%** (**180/344**) | **72.7%** (**250/344**) |

Tokenized with Gemma's SentencePiece BPE tokenizer, this corresponds to 370–425 BPE tokens (mean 394 tokens, Table 3), consuming 11 to 12 percent of the 3,480-token prompt budget and leaving over 87 percent open for model reflection and generation. Paralleling the retrieval evaluation in Section 5.4, standalone Perron matches BM25 in function inclusion (51.2%, 176/344 vs. 51.5%, 177/344, a one-instance difference of −0.29 percentage points), while lexical search preserves higher file-level grounding (72.4% vs. 67.2%). Table 3 inclusion counts credit methods within enclosing class breadcrumb scopes (yielding 177 and 176 hits vs. 175 and 173 in Table 2). Allocating a hard slot budget (8 lexical slots and 2 structural diffusion slots) achieves 52.3% function inclusion (180/344) and 72.7% file inclusion (250/344) at an extraction latency of 513.96 ms (measuring joint pipeline execution, where sparse CSR diffusion requires only 1.89 ms on Requests and 5.10 ms on SymPy, while AST symbol parsing and IO account for the remainder).

## 5.6 Ablation Study

To evaluate the contribution of individual mechanisms under controlled conditions:

- **Without Specificity Ratio** ($\gamma = 0$): Perron at $\gamma = 0$ reduces identically to Standard PPR. Under diffuse linear teleportation on raw graphs ($N = 237$), unweighted diffusion collapses Function Acc@10 from 34.2% (81/237) to 10.1% (24/237, $\Delta = -24.1$ pp) as test-node contamination rises to 32.5%. On SWE-bench Verified ($N = 344$), removing specificity weighting drops HSI@10 from 99.5% to 90.8% (introducing 9.2% hub contamination), while top-10 accuracy remains at 50.3% vs. 50.0%.
- **Without Graph Sanitization (Raw vs. Sanitized)**: Retaining un-sanitized test subgraphs under diffuse priors funnels random-walk mass into testing harnesses, reducing Function Acc@10 by 3.8 pp (13.9% vs. 10.1%). Concentrating the teleport prior via top-25 softmax eliminates test-sink vulnerability even without structural pruning.

## 6 Open-Source Ecosystem and Resource Deliverables

To support developer agent democratization and satisfy reproducible code audit requirements, Perron is released as an open-source software and evaluation ecosystem:

- **`perron-core` v0.3.0**: a Python package (`numpy`, `scipy`, `pyyaml`) providing zero-copy memory-mapped CSR ingestion, prior normalization, query-directed specificity diffusion, and AST editing, published on PyPI at https://pypi.org/project/perron-core/0.3.0/.
- **MCP server (`perron-mcp`)**: a JSON-RPC 2.0 stdio server (`perron.mcp_server`) for desktop agent environments with tools `retrieve_context`, `inspect_symbol_breadcrumbs`, and `build_code_graph`.
- **Dataset `swebench-lite-codegraphs`**: CSR edge topologies, dangling indicators, and stationary distributions for all 12 SWE-bench Lite repositories under Apache 2.0.
- **Dataset `swebench-strata-annotations`**: 800 per-instance lexical annotations across SWE-bench Lite and Verified (707 unique instances, 93 overlap) under CC0-1.0 at https://www.kaggle.com/datasets/yvliet/swebench-strata-annotations.
- **Kaggle Gemma 4 Workstation Kernel**: an interactive GPU demonstration notebook at https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation.

## 7 Discussion and Limitations

While Perron acts as an invariant topological stabilizer under noisy priors and sustains robust hub suppression (99.5% HSI), several limitations remain: (1) static call graphs miss dynamic runtime dispatch, reflection (`getattr`), or monkey-patching; (2) across disconnected modules, diffusion relies entirely on the teleport prior; (3) structural diffusion alone does not outperform lexical BM25 on issue descriptions that explicitly quote target identifiers; (4) while reader transmission was benchmarked on Gemma 4 E2B ($N = 100$), unconstrained generative patch synthesis was not evaluated, as small-model reflection loops require higher generation budgets than the 256-token transmission envelope; and (5) the reference engine targets Python AST; while an initial polyglot Tree-sitter interface smoke test is implemented for TypeScript (`perron/polyglot/treesitter.py`) to verify CSR protocol extensibility, large-scale multilingual SWE-bench evaluation remains an open research direction.

## 8 Conclusion

On scale-free code graphs, Personalized PageRank collapse under default damping ($\beta = 0.85$) is driven by teleport prior dispersion and test-node sink absorption. Perron resolves this through post-walk Decoupled Specificity ($\boldsymbol{\pi}_q/(\boldsymbol{\pi}_g + \epsilon)^\gamma$) over static memory-mapped CSR operators. In a pre-registered confirmatory evaluation on SWE-bench Verified (344 callable tasks), Perron exhibits no statistically detectable difference from BM25 at top-10 (50.3% vs. 50.9%, non-inferiority inconclusive against a $-2.0$ pp margin). While their theoretical oracle union reaches 54.4% (187/344), unrouted rank fusion fails to surpass standalone BM25, demonstrating that disjoint hits reflect top-10 boundary rank permutations rather than a systematic orthogonal structural discovery mechanism. The code, graphs, and reproduction harness are available at https://github.com/yvliet/perron and https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation.

## References

[1] Reid Andersen, Fan Chung, and Kevin Lang. Local graph partitioning using PageRank vectors. In *Proceedings of the 47th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 475–486, 2006.

[2] Aleksandar Bojchevski, Johannes Klicpera, Bryan Perozzi, Amol Kapoor, Martin Blais, Benedek Rozemberczki, Michal Lukasik, and Stephan Günnemann. Scaling graph neural networks with approximate PageRank. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 2464–2473, 2020.

[3] Alfred Brauer. Limits for the characteristic roots of a matrix. IV: Applications to stochastic matrices. *Duke Mathematical Journal*, 19(1):75–91, 1952.

[4] Zhaoling Chen, Robert Tang, Gangda Deng, Fang Wu, Jialong Wu, Zhiwei Jiang, Viktor K. Prasanna, Arman Cohan, and Xingyao Wang. LocAgent: Graph-guided LLM agents for code localization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[5] Paul Gauthier. Aider: Ai pair programming in your terminal. https://github.com/paul-gauthier/aider, 2023.

[6] Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[7] Taher H. Haveliwala and Sepandar D. Kamvar. The second eigenvalue of the Google matrix. *Stanford University Technical Report*, 2003.

[8] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

[9] Johannes Klicpera, Aleksandar Bojchevski, and Stephan Günnemann. Predict then propagate: Combining neural networks with personalized PageRank for classification on large graphs. In *International Conference on Learning Representations (ICLR)*, 2019.

[10] Omer Levy and Yoav Goldberg. Neural word embedding as implicit matrix factorization. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27, pages 2177–2185, 2014.

[11] David Liben-Nowell and Jon Kleinberg. The link-prediction problem for social networks. *Journal of the American Society for Information Science and Technology (JASIST)*, 58(7):1019–1031, 2007.

[12] Xiangyan Liu, Bo Lan, Zhiyuan Hu, Yang Liu, Zhicheng Zhang, Fei Wang, Michael Shieh, and Wenmeng Zhou. CodexGraph: Bridging large language models and code repositories via code graph databases. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.

[13] Tomas Mikolov, Ilya Sutskever, Kai Chen, Gregory S. Corrado, and Jeffrey Dean. Distributed representations of words and phrases and their compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 26, pages 3111–3119, 2013.

[14] Siru Ouyang, Wenhao Yu, Kaixin Ma, Zilin Xiao, Zhihan Zhang, Mengzhao Jia, Jiawei Han, Hongming Zhang, and Dong Yu. RepoGraph: Enhancing AI software engineering with repository-level code graph. In *International Conference on Learning Representations (ICLR)*, 2025.

[15] Spencer Pearson, José Campos, René Just, Gordon Fraser, Rui Abreu, Michael D. Ernst, Deric Pang, and Benjamin Keller. Evaluating and improving fault localization techniques. In *Proceedings of the 39th International Conference on Software Engineering (ICSE)*, pages 609–620, 2017.

[16] Mark Weiser. Program slicing. *IEEE Transactions on Software Engineering*, SE-10(4):352–357, 1984.

[17] Chunqiu Steven Xia, Yinlin Deng, Soren Dunn, and Lingming Zhang. Agentless: Demystifying LLM-based software engineering agents. In *Proceedings of the 33rd ACM SIGSOFT International Symposium on the Foundations of Software Engineering (FSE)*, 2025.

[18] John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 50528–50652, 2024.

[19] Fengji Zhang, Bei Chen, Yue Zhang, Jin Liu, Daoguang Zan, Yi Mao, Jian-Guang Lou, and Lingpeng Kong. RepoCoder: Repository-level code completion through iterative retrieval and generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2471–2484, 2023.

[20] Yuntong Zhang, Zhiyu Li, Min Lin, Denghui Zhang, Hongyu Zhang, and Lingxiao Jiang. AutoCodeRover: Autonomous program improvement. In *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, pages 762–774, 2024.

[21] Tao Zhou, Zoltán Kuscsik, Jian-Guo Liu, Matúš Medo, Joseph R. Wakeling, and Yi-Cheng Zhang. Solving the apparent diversity-accuracy dilemma of recommender systems. *Proceedings of the National Academy of Sciences (PNAS)*, 107(10):4511–4515, 2010.

[22] Daming Zou, Jingjing Liang, Yuyan Xiong, Michael D. Ernst, and Lu Zhang. An empirical study of fault localization families and their combinations. *IEEE Transactions on Software Engineering*, 47(2):332–347, 2021.

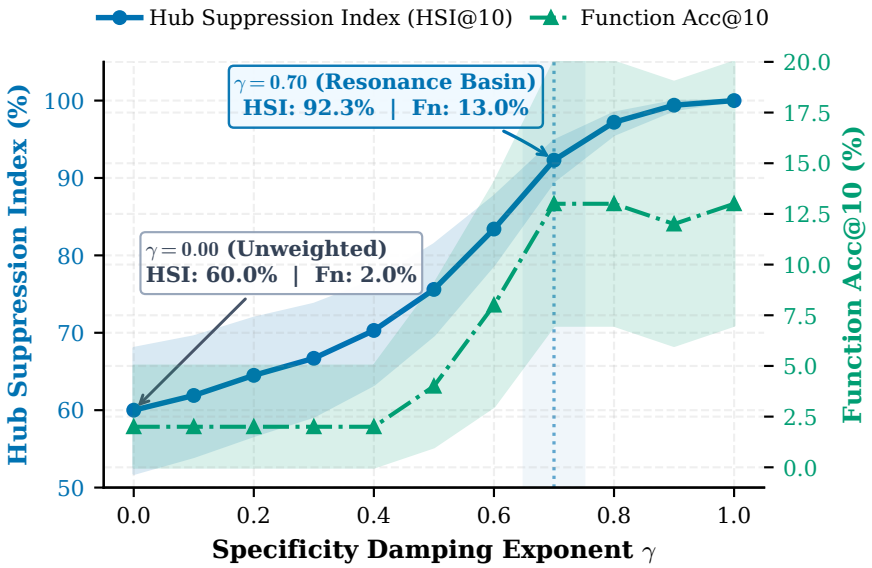



**Figure 4: Specificity exponent sweep on dev_val ($N = 100$).** Function Acc@10 and HSI@10 as a function of $\gamma$; $\gamma = 0$ is unweighted PPR. At the default $\gamma = 0.70$: Function Acc@10 13.0%, HSI@10 92.3%; at $\gamma = 0$: 2.0% and 60.0%.

# A Supplementary Retrieval Tables

To evaluate retrieval behavior across varying degrees of textual specificity, Table 4 stratifies performance on SWE-bench Verified ($N = 344$ callable tasks) into three lexical accessibility strata: Stratum A where the gold target symbol is explicitly named in the problem description, Stratum B where the gold symbol is a direct neighbor of an issue mention, and Stratum C where the gold symbol is structurally distant without textual mention. Furthermore, Figure 4 plots the sensitivity of Function Acc@10 and HSI@10 across specificity exponent $\gamma \in [0, 1]$ on the development split ($N = 100$, sampled from SWE-bench Lite, disjoint from held-out evaluations).

**Table 4: Retrieval performance stratified by lexical accessibility on SWE-bench Verified** ($N = 344$ callable tasks). Stratum A: gold target symbol is explicitly referenced in the issue description ($N = 208$). Stratum B: gold symbol is 1-hop neighbor of an issue mention ($N = 5$). Stratum C: gold symbol is structurally distant without direct mention in the issue text ($N = 131$). Top-10 hits and accuracy rates are reported.

| **Method** | **Stratum A** ($N = 208$) | **Stratum B** ($N = 5$) | **Stratum C** ($N = 131$) | **Overall** ($N = 344$) |
|---|---|---|---|---|
| Okapi BM25 | 147 (70.7%) | 0 (0.0%) | 28 (21.4%) | 175 (50.9%) |
| Standard PPR ($\beta = 0.85$) | 147 (70.7%) | 0 (0.0%) | 25 (19.1%) | 172 (50.0%) |
| Degree-Normalized PPR | 146 (70.2%) | 0 (0.0%) | 24 (18.3%) | 170 (49.4%) |
| HippoRAG Prior Scaling | 142 (68.3%) | 0 (0.0%) | 28 (21.4%) | 170 (49.4%) |
| Perron Static ($\gamma = 0.70$) | 146 (70.2%) | 0 (0.0%) | 27 (20.6%) | 173 (50.3%) |
| Perron Adaptive | 146 (70.2%) | 0 (0.0%) | 27 (20.6%) | 173 (50.3%) |
| Slot Fusion (RRF) | 147 (70.7%) | 0 (0.0%) | 27 (20.6%) | 174 (50.6%) |